Automated Guided Vehicle Scheduling Problem

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Inhalt

- Ausgangslage
- Ziel
- Beschränkungen
- Decision Support System

Ausgangslage

- Gegeben:
  - Menge von Transferaufträgen
  - Anzahl AGVs (automated guided vehicles)
  - Energieverbrauch pro Auftrag
  - Bearbeitungszeit pro Auftrag
  - Batteriekapazität und Ladezeit

Ziel

- Gesucht:
  - Zeitplan mit minimaler Gesamtdurchlaufzeit
  - Einhaltung Batteriekapazität

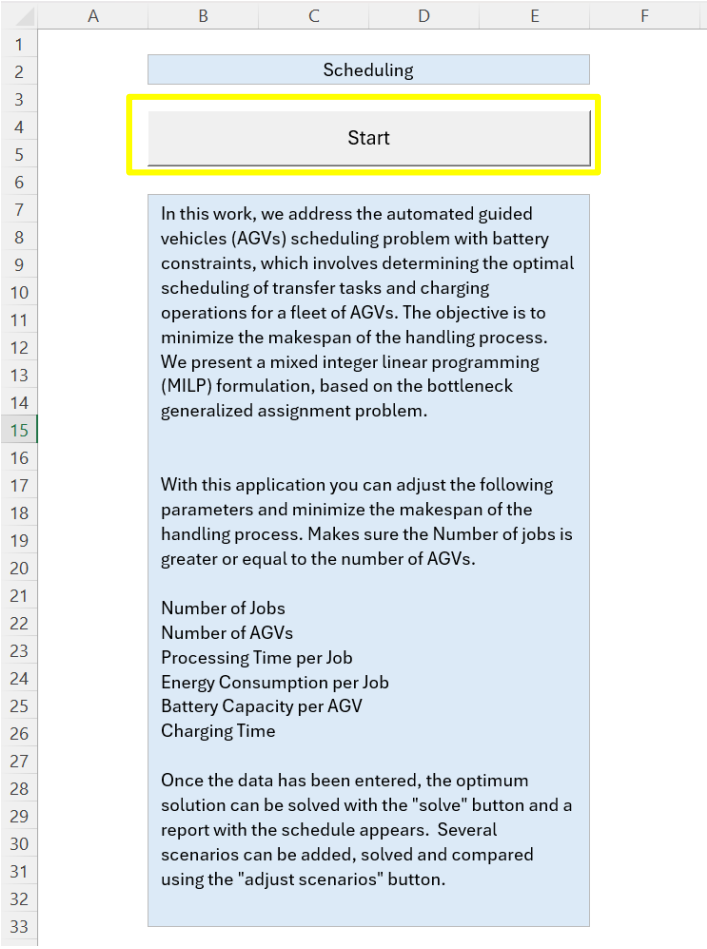
Beschränkungen

- Jeder Transferauftrag muss ausgeführt werden
- Energieverbrauch
- Energiereserve

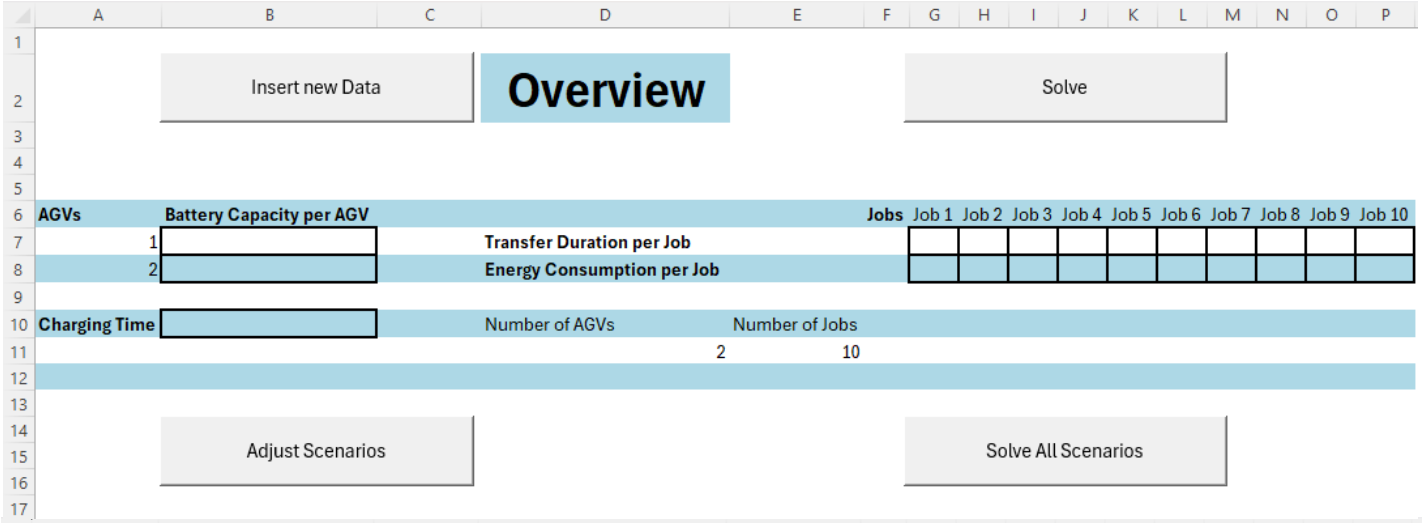
Decision Support System

- Vorteile von unserem System
  - Effizienz
  - Ablaufplanung
  - Vergleich von verschiedenen Fahrzeugen
  - Visualisierung

Beispielfolie «Start»



Beispielfolie «Overview»



Insert new Data

Overview

Solve

AGVs	Battery Capacity per AGV
1	
2	

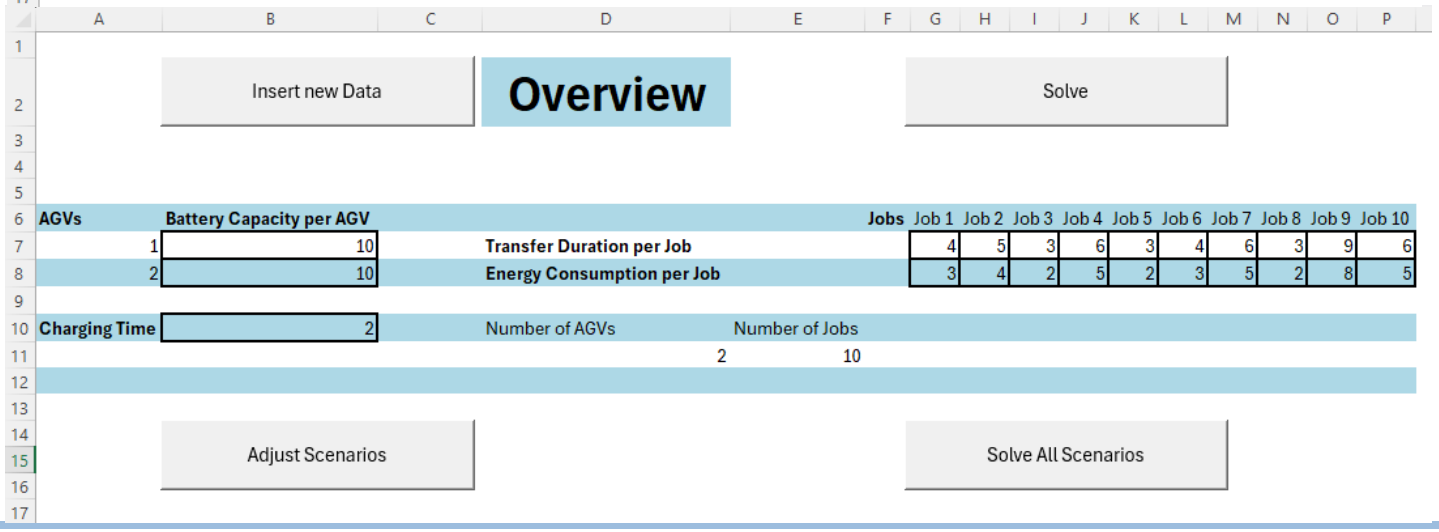
Jobs	Job 1	Job 2	Job 3	Job 4	Job 5	Job 6	Job 7	Job 8	Job 9	Job 10
Transfer Duration per Job										
Energy Consumption per Job										

Charging Time	

Number of AGVs	Number of Jobs
2	10

Adjust Scenarios

Solve All Scenarios



Insert new Data

Overview

Solve

AGVs	Battery Capacity per AGV
1	10
2	10

Jobs	Job 1	Job 2	Job 3	Job 4	Job 5	Job 6	Job 7	Job 8	Job 9	Job 10
Transfer Duration per Job	4	5	3	6	3	4	6	3	9	6
Energy Consumption per Job	3	4	2	5	2	3	5	2	8	5

Charging Time	2

Number of AGVs	Number of Jobs
2	10

Adjust Scenarios

Solve All Scenarios

Beispielfolie «Report 1/3»

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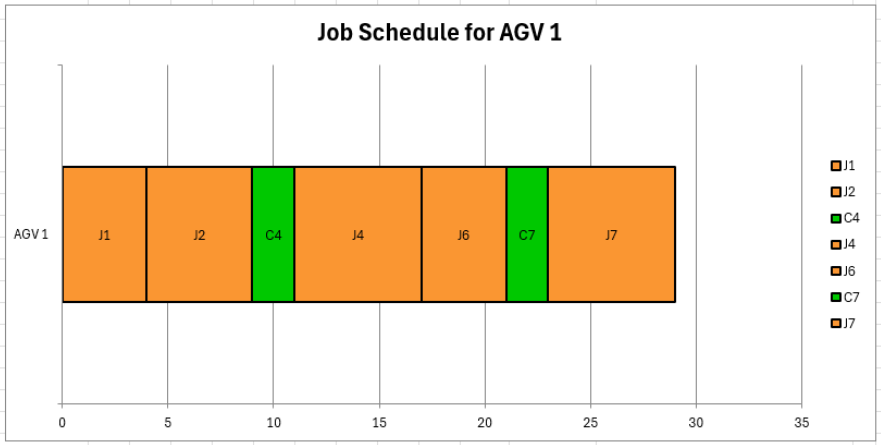
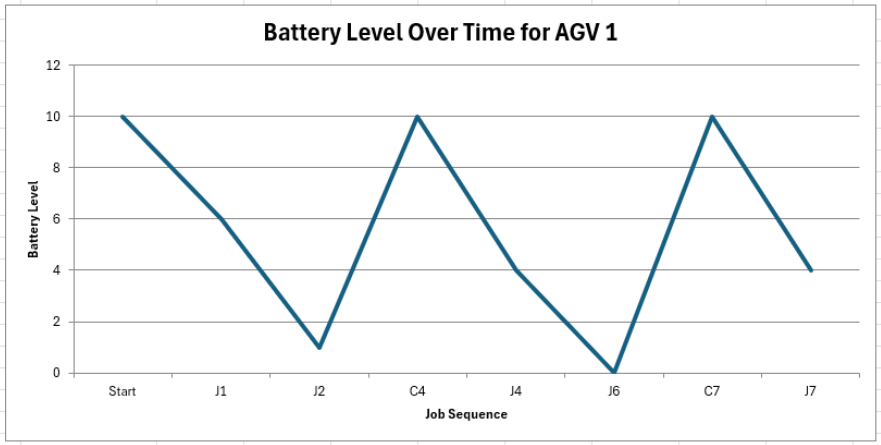
	A	B	C	D	E	F	G	H	I	J	K
1	Transfer Job duration	Job 1	Job 2	Job 3	Job 4	Job 5	Job 6	Job 7	Job 8	Job 9	Job 10
2	AGV 1	4	5	3	6	3	4	6	3	9	6
3	AGV 2	4	5	3	6	3	4	6	3	9	6
4	Charging Job duration	Job 1	Job 2	Job 3	Job 4	Job 5	Job 6	Job 7	Job 8	Job 9	Job 10
5	AGV 1	2	2	2	2	2	2	2	2	2	2
6	AGV 2	2	2	2	2	2	2	2	2	2	2
7	Energy Consumption	Job 1	Job 2	Job 3	Job 4	Job 5	Job 6	Job 7	Job 8	Job 9	Job 10
8	AGV 1	4	5	3	6	3	4	6	2	8	5
9	AGV 2	4	5	3	6	3	4	6	2	8	5
10											

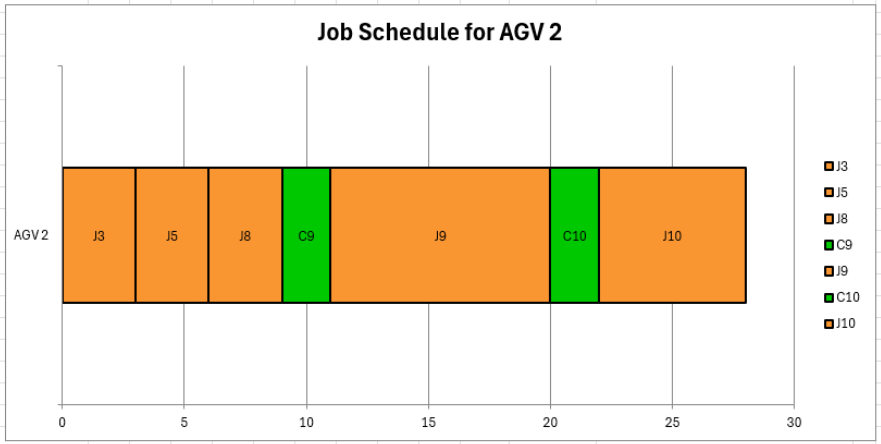
	M	N
	Battery Capacity per AGV	
	AGV 1	10
	AGV 2	10
	Charging Time	2
	Completion Time of Transfer Process	29

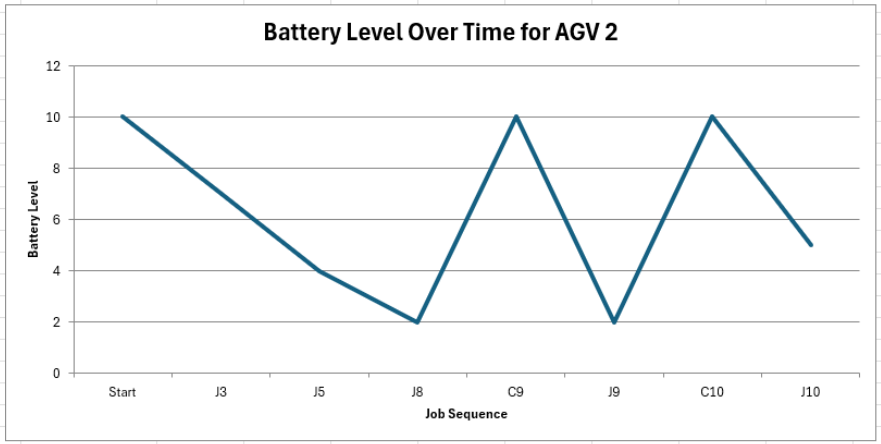
Report

Beispielfolie «Report 2/3»









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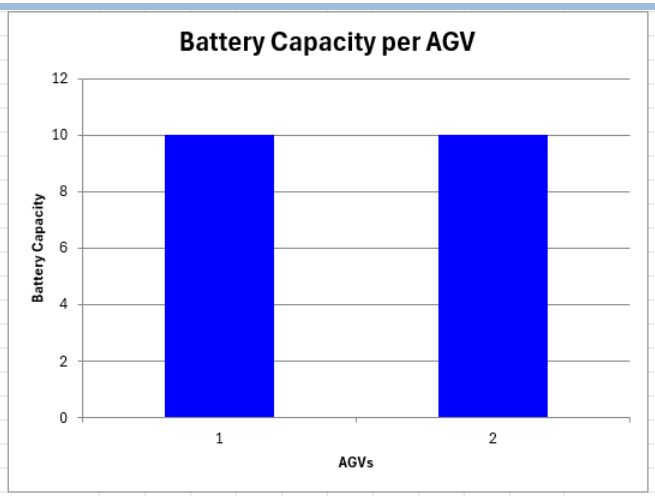
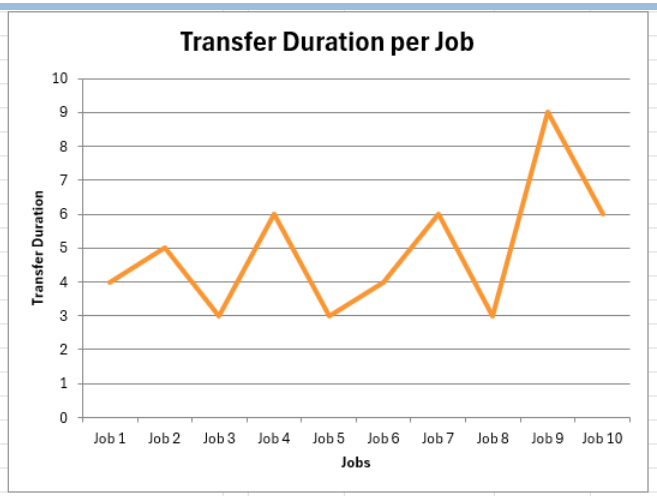
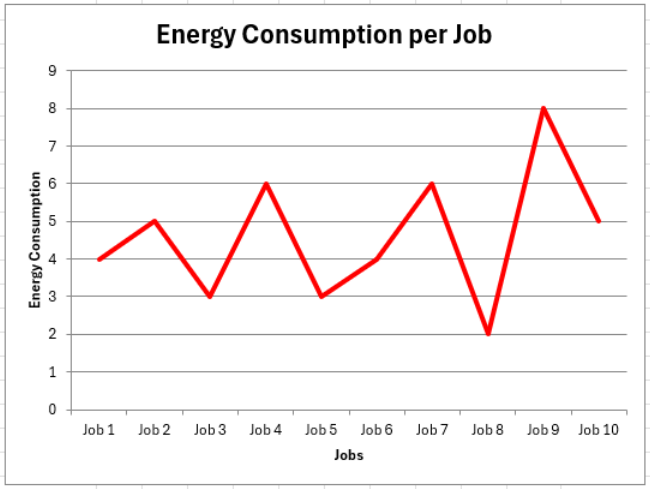
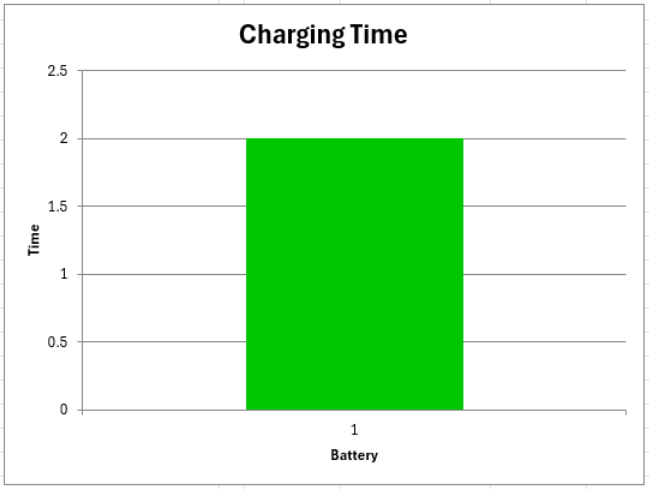








Beispielfolie «Adjust Scenarios 1/2»

Insert new Data

Overview

Solve

AGVs	Battery Capacity per AGV
1	10
2	10

Charging Time	2

Jobs	Job 1	Job 2	Job 3	Job 4	Job 5	Job 6	Job 7	Job 8	Job 9	Job 10
Transfer Duration per Job	4	5	3	6	3	4	6	3	9	6
Energy Consumption per Job	4	5	3	6	3	4	6	2	8	5

Number of AGVs	Number of Jobs
2	10

Number of Scenarios

4

Adjust Scenarios

Solve All Scenarios

AGVs Scenario 1	Battery Capacity per AGV
1	
2	

AGVs Scenario 2	Battery Capacity per AGV
1	
2	
3	

AGVs Scenario 3	Battery Capacity per AGV
1	
2	
3	
4	

AGVs Scenario 4	Battery Capacity per AGV
1	
2	
3	
4	
5	

Beispielfolie «Adjust Scenarios 2/2»

Insert new Data

Overview

Solve

AGVs	Battery Capacity per AGV
1	10
2	10

Jobs	Job 1	Job 2	Job 3	Job 4	Job 5	Job 6	Job 7	Job 8	Job 9	Job 10
Transfer Duration per Job	4	5	3	6	3	4	6	3	9	6
Energy Consumption per Job	4	5	3	6	3	4	6	2	8	5

Charging Time	2

Number of AGVs	Number of Jobs
2	10

Number of Scenarios
4

Adjust Scenarios

Solve All Scenarios

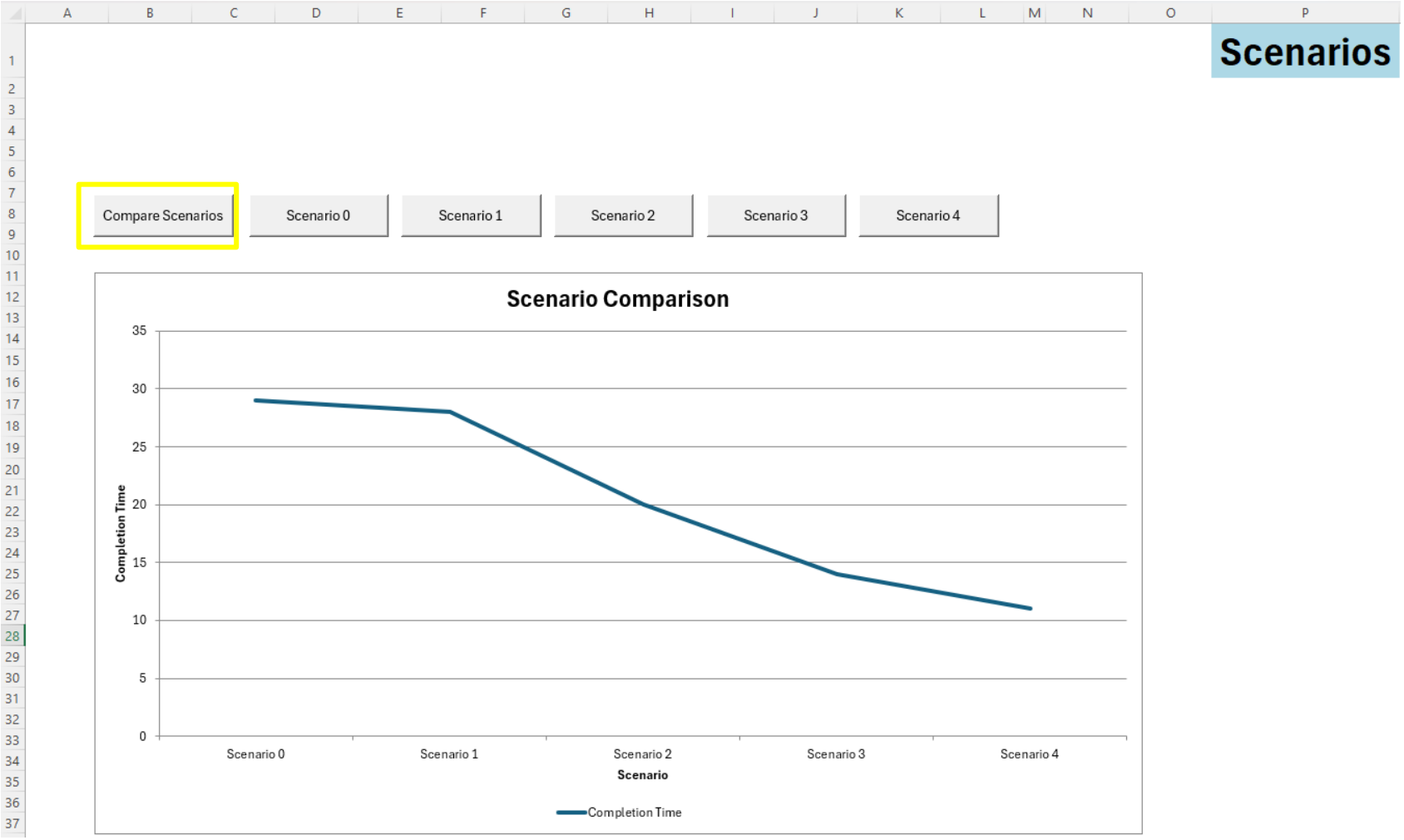
AGVs Scenario 1	Battery Capacity per AGV
1	12
2	8

AGVs Scenario 2	Battery Capacity per AGV
1	10
2	6
3	6

AGVs Scenario 3	Battery Capacity per AGV
1	11
2	6
3	8
4	9

AGVs Scenario 4	Battery Capacity per AGV
1	10
2	10
3	8
4	8
5	6

Beispielfolie «Scenarios»



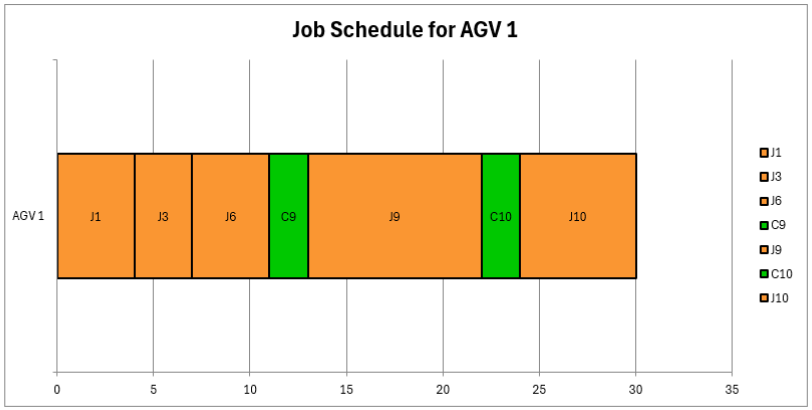
Beispielfolie «Scenario 1, 1/3»

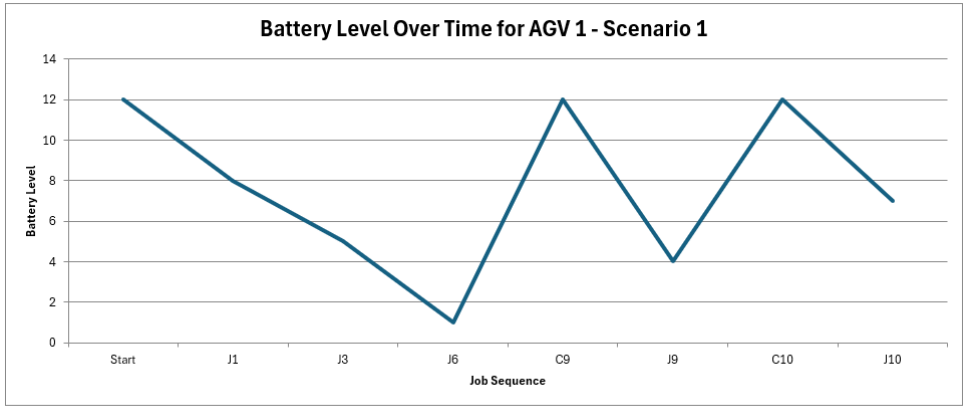
Compare Scenarios | Scenario 0 | Scenario 1 | Scenario 2 | Scenario 3 | Scenario 4

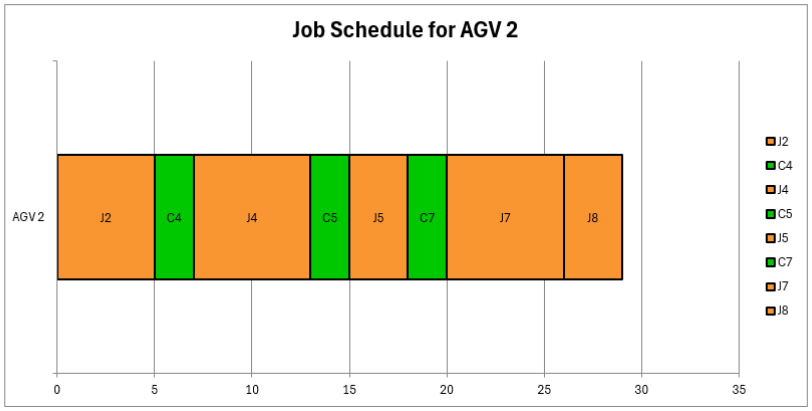
Transfer Job	Job 1	Job 2	Job 3	Job 4	Job 5	Job 6	Job 7	Job 8	Job 9	Job 10
AGV 1	4	5	3	6	3	4	6	3	9	6
AGV 2	4	5	3	6	3	4	6	3	9	6
Charging Job	Job 1	Job 2	Job 3	Job 4	Job 5	Job 6	Job 7	Job 8	Job 9	Job 10
AGV 1	2	2	2	2	2	2	2	2	2	2
AGV 2	2	2	2	2	2	2	2	2	2	2
Energy Consu	Job 1	Job 2	Job 3	Job 4	Job 5	Job 6	Job 7	Job 8	Job 9	Job 10
AGV 1	4	5	3	6	3	4	6	2	8	5
AGV 2	4	5	3	6	3	4	6	2	8	5

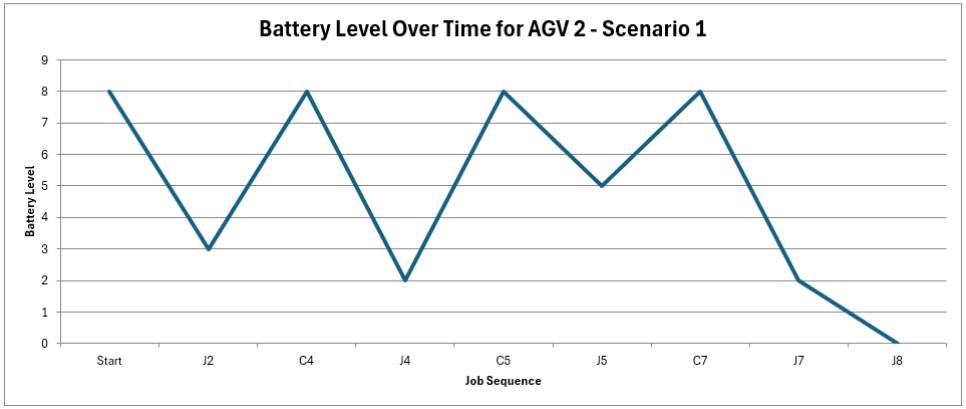
Battery Capacity per AGV	
AGV 1	12
AGV 2	8
Charging Tim	2
Completion T	28

Beispielfolie «Scenario 1, 2/3»

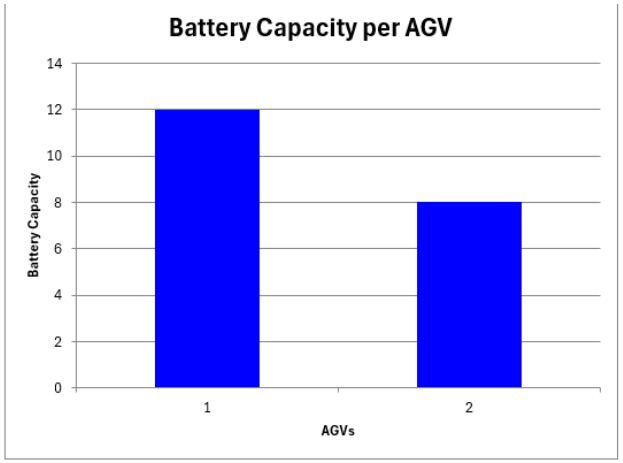


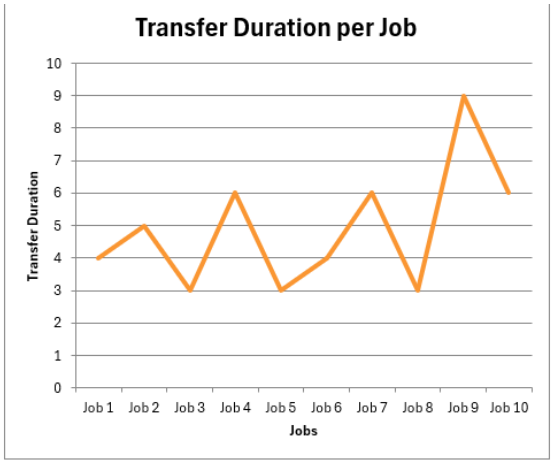


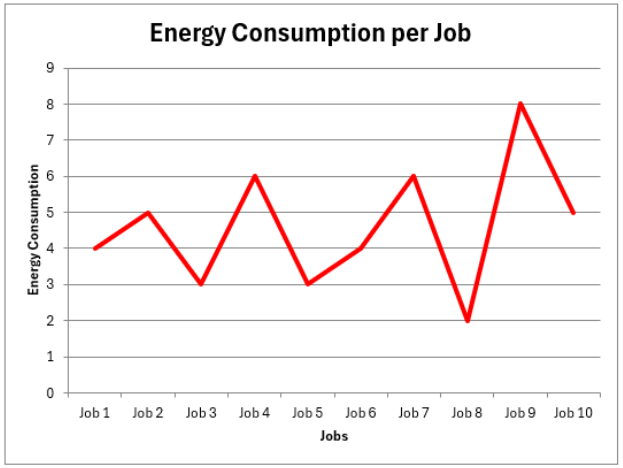


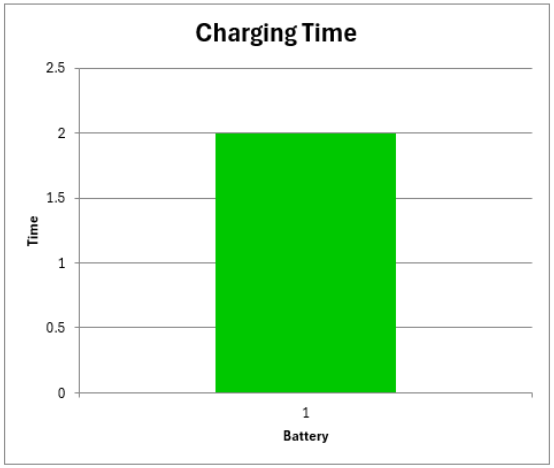


Beispielfolie «Scenario 1, 3/3»









Beispielfolie «Scenario 2, 1/4»

Compare Scenarios | Scenario 0 | Scenario 1 | Scenario 2 | Scenario 3 | Scenario 4

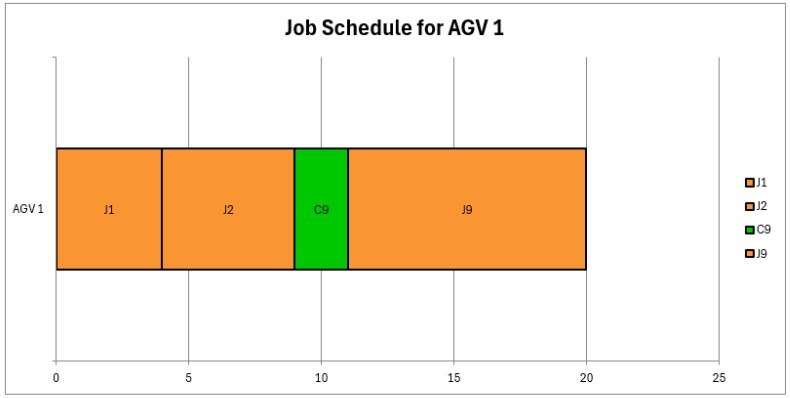
Transfer Job	Job 1	Job 2	Job 3	Job 4	Job 5	Job 6	Job 7	Job 8	Job 9	Job 10
AGV 1	4	5	3	6	3	4	6	3	9	6
AGV 2	4	5	3	6	3	4	6	3	9	6
AGV 3	4	5	3	6	3	4	6	3	9	6

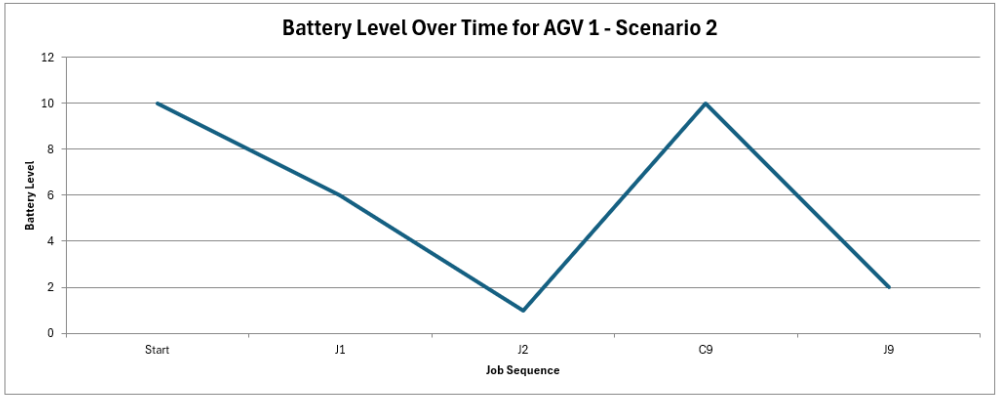
Charging Job	Job 1	Job 2	Job 3	Job 4	Job 5	Job 6	Job 7	Job 8	Job 9	Job 10
AGV 1	2	2	2	2	2	2	2	2	2	2
AGV 2	2	2	2	2	2	2	2	2	2	2
AGV 3	2	2	2	2	2	2	2	2	2	2

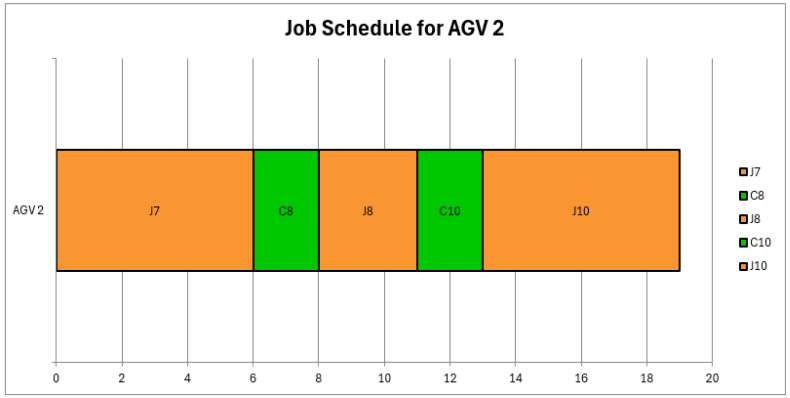
Energy Consu	Job 1	Job 2	Job 3	Job 4	Job 5	Job 6	Job 7	Job 8	Job 9	Job 10
AGV 1	4	5	3	6	3	4	6	2	8	5
AGV 2	4	5	3	6	3	4	6	2	8	5
AGV 3	4	5	3	6	3	4	6	2	8	5

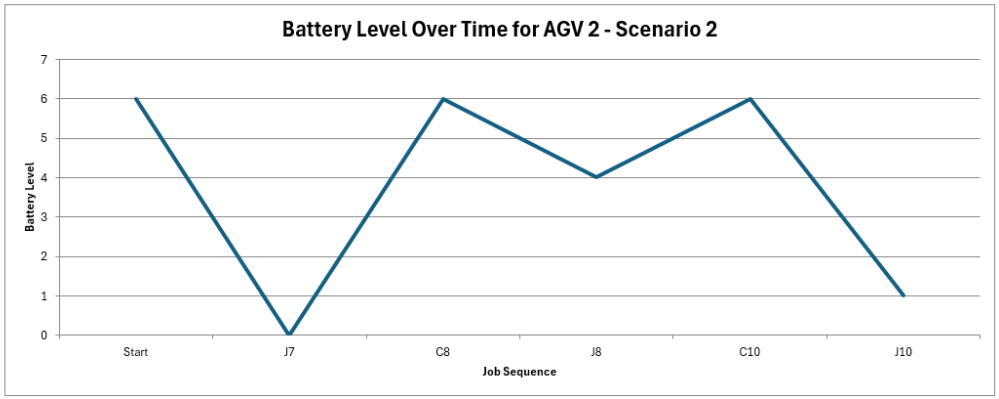
Battery Capacity per AGV	
AGV 1	10
AGV 2	6
AGV 3	6
Charging Tim	2
Completion T	20

Beispielfolie «Scenario 2, 2/4»









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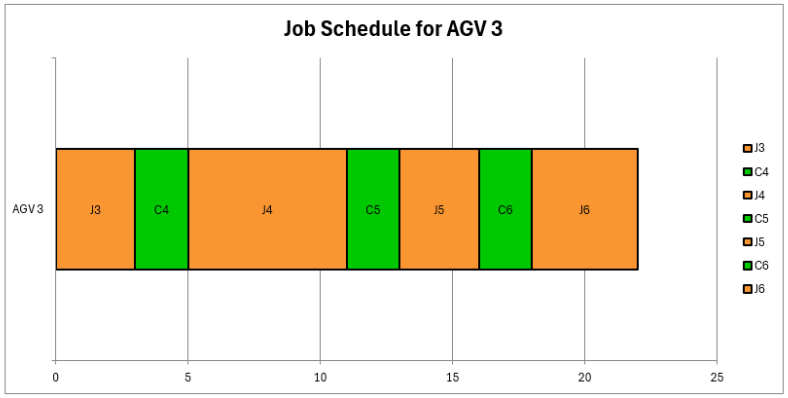

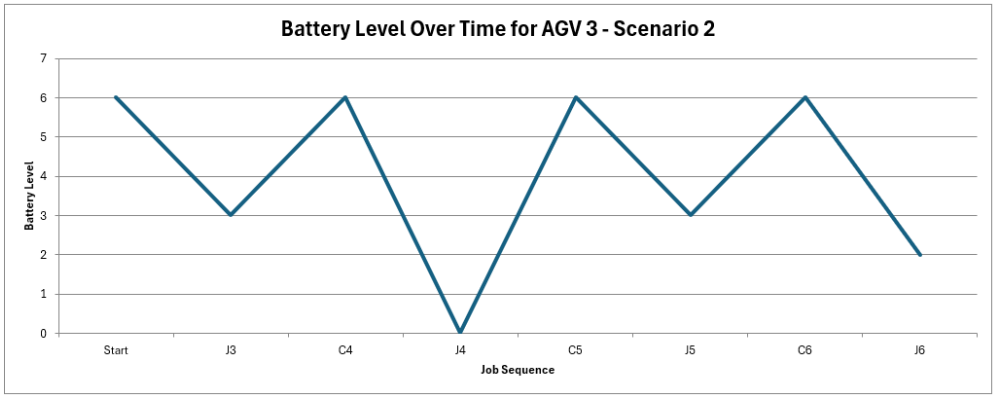

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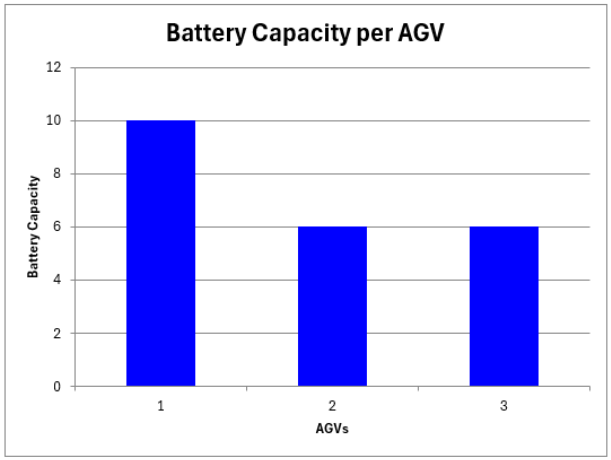
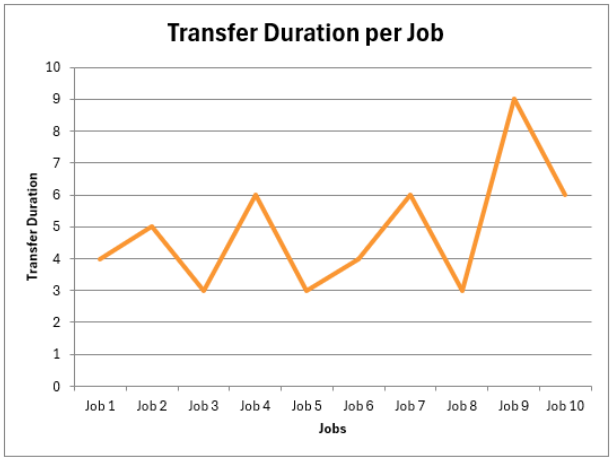
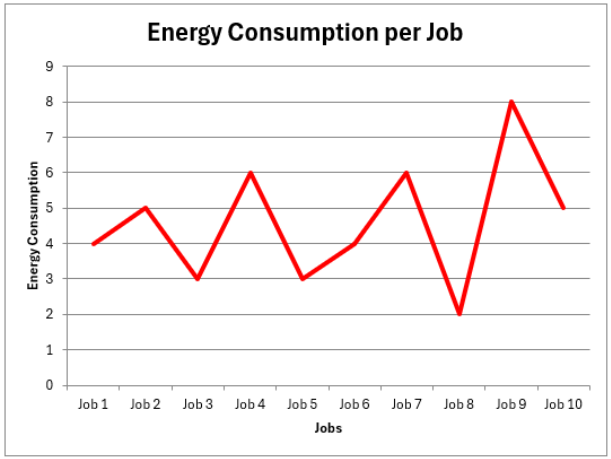
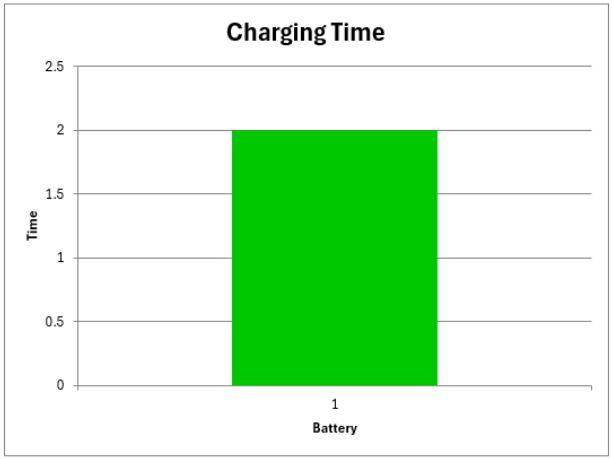








Beispielfolie «Scenario 3, 1/4»

Compare Scenarios | Scenario 0 | Scenario 1 | Scenario 2 | Scenario 3 | Scenario 4

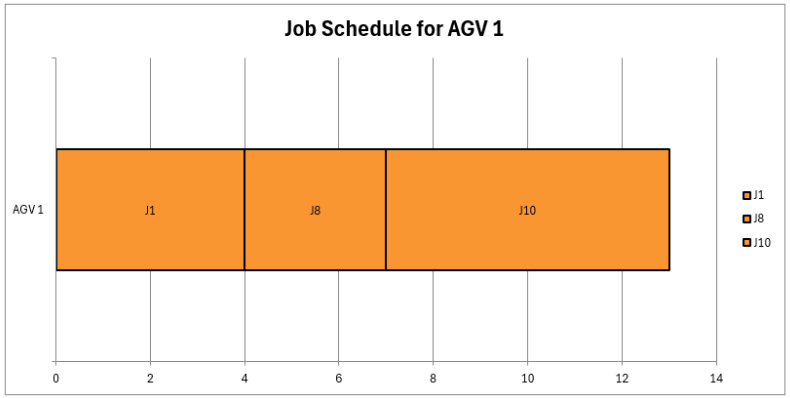
Transfer Job	Job 1	Job 2	Job 3	Job 4	Job 5	Job 6	Job 7	Job 8	Job 9	Job 10
AGV 1	4	5	3	6	3	4	6	3	9	6
AGV 2	4	5	3	6	3	4	6	3	9	6
AGV 3	4	5	3	6	3	4	6	3	9	6
AGV 4	4	5	3	6	3	4	6	3	9	6

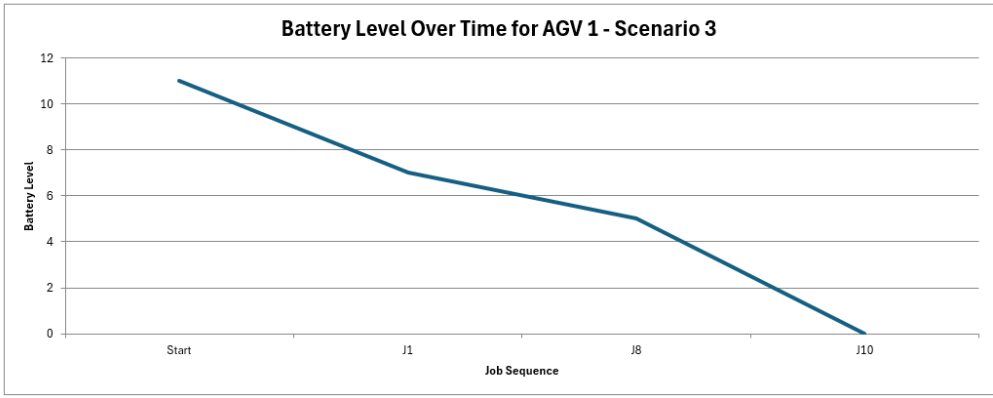
Charging Job	Job 1	Job 2	Job 3	Job 4	Job 5	Job 6	Job 7	Job 8	Job 9	Job 10
AGV 1	2	2	2	2	2	2	2	2	2	2
AGV 2	2	2	2	2	2	2	2	2	2	2
AGV 3	2	2	2	2	2	2	2	2	2	2
AGV 4	2	2	2	2	2	2	2	2	2	2

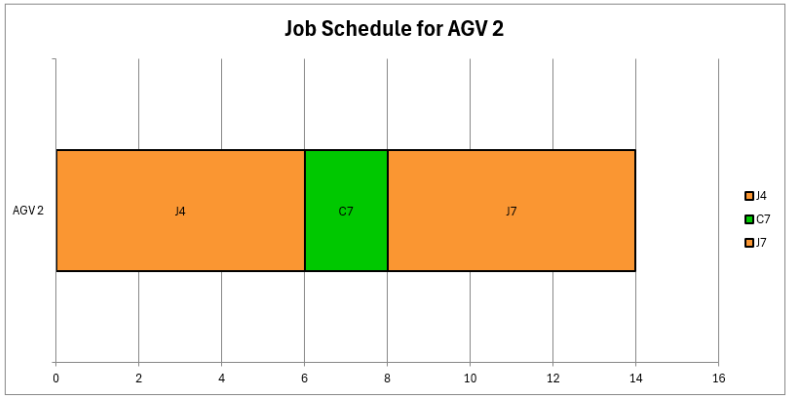
Energy Consu	Job 1	Job 2	Job 3	Job 4	Job 5	Job 6	Job 7	Job 8	Job 9	Job 10
AGV 1	4	5	3	6	3	4	6	2	8	5
AGV 2	4	5	3	6	3	4	6	2	8	5
AGV 3	4	5	3	6	3	4	6	2	8	5
AGV 4	4	5	3	6	3	4	6	2	8	5

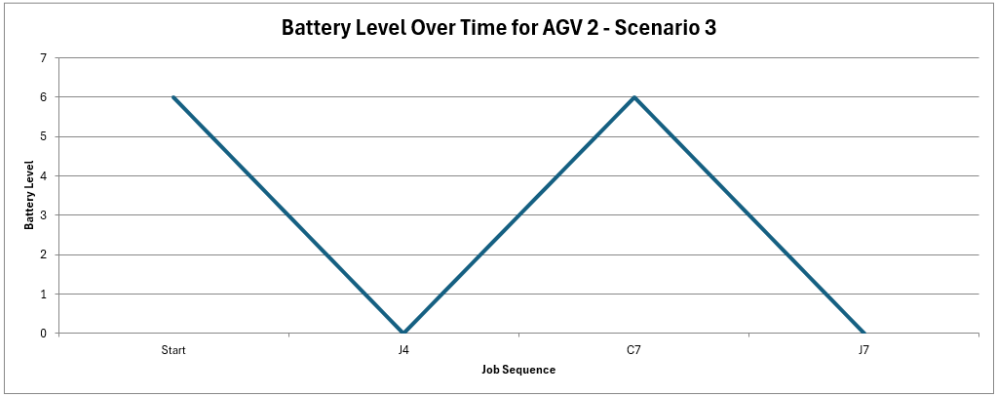
Battery Capacity per AGV	
AGV 1	11
AGV 2	6
AGV 3	8
AGV 4	9
Charging Tim	2
Completion T	14

Beispielfolie «Scenario 3, 2/4»

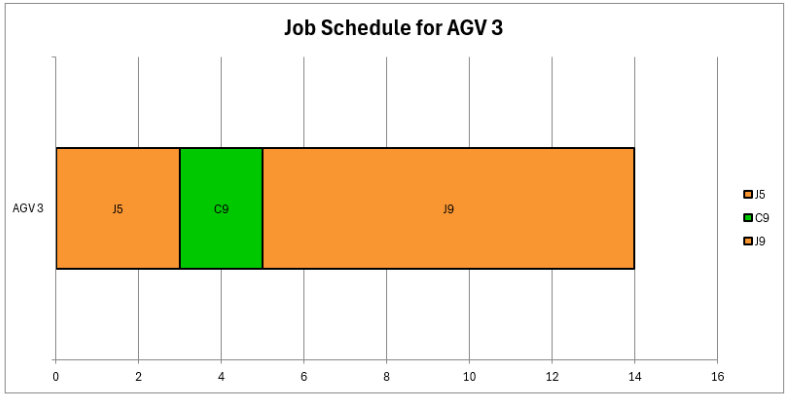


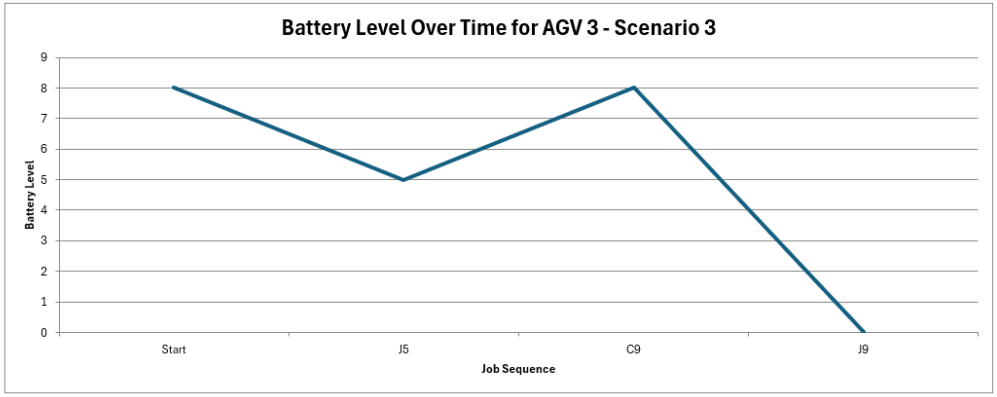


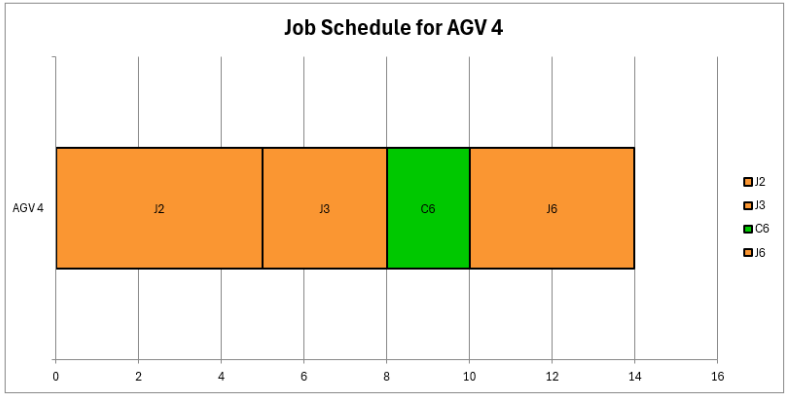


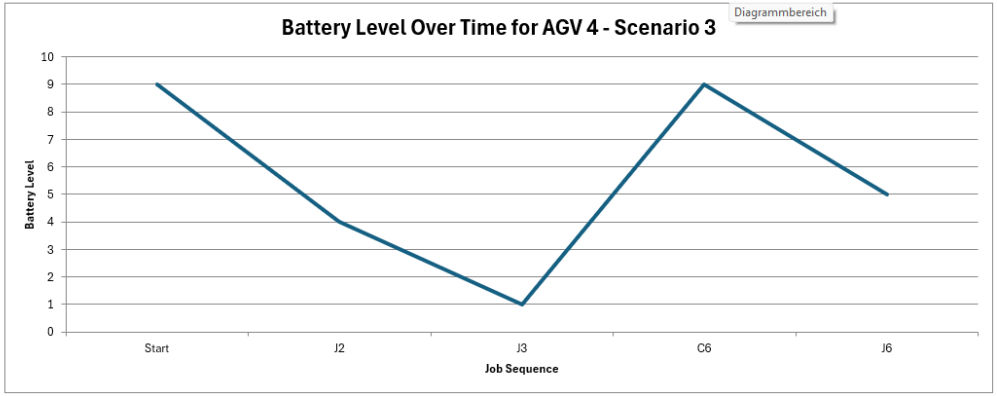


Beispielfolie «Scenario 3, 3/4»

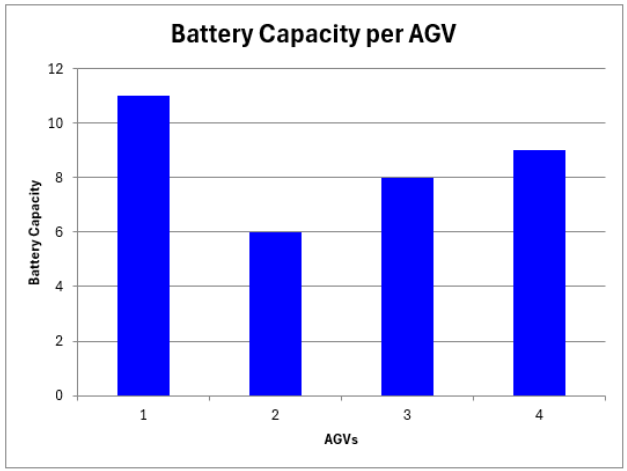


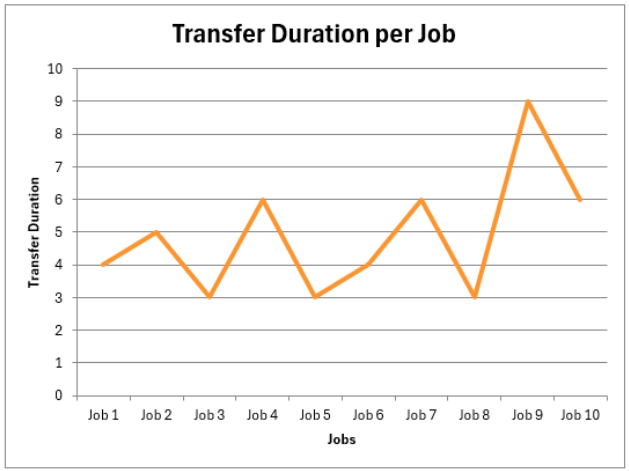


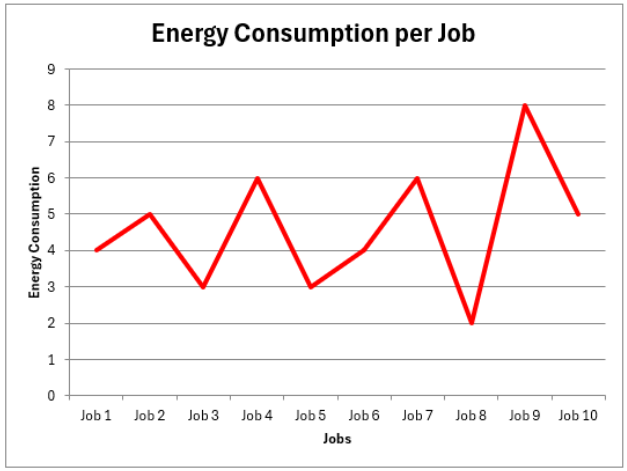


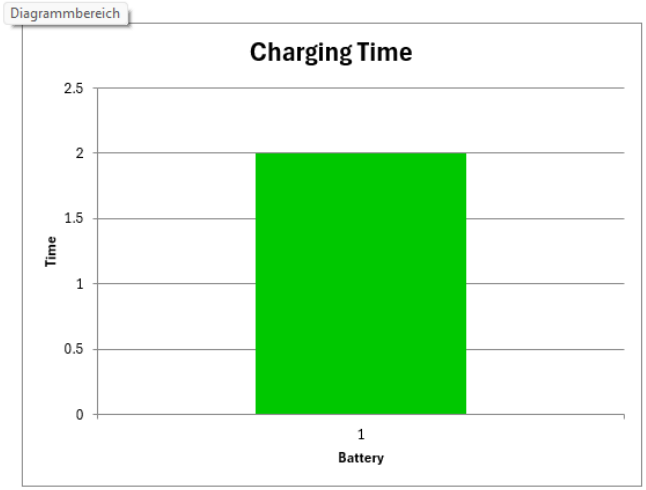


Beispielfolie «Scenario 3, 4/4»









Beispielfolie «Scenario 4, 1/5»

Compare Scenarios | Scenario 0 | Scenario 1 | Scenario 2 | Scenario 3 | Scenario 4

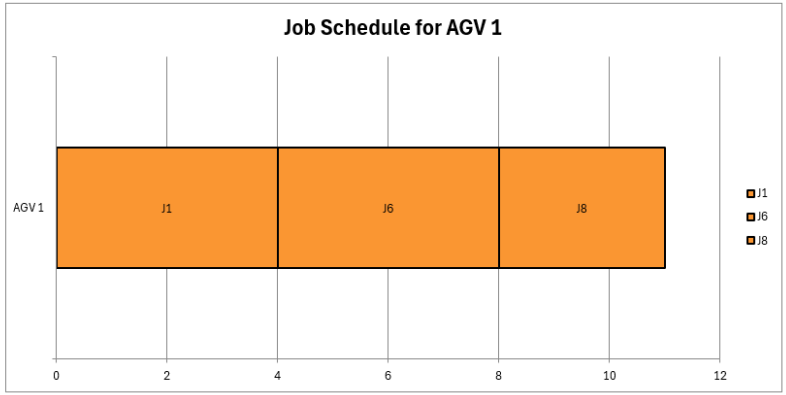
Transfer Job	Job 1	Job 2	Job 3	Job 4	Job 5	Job 6	Job 7	Job 8	Job 9	Job 10
AGV 1	4	5	3	6	3	4	6	3	9	6
AGV 2	4	5	3	6	3	4	6	3	9	6
AGV 3	4	5	3	6	3	4	6	3	9	6
AGV 4	4	5	3	6	3	4	6	3	9	6
AGV 5	4	5	3	6	3	4	6	3	9	6

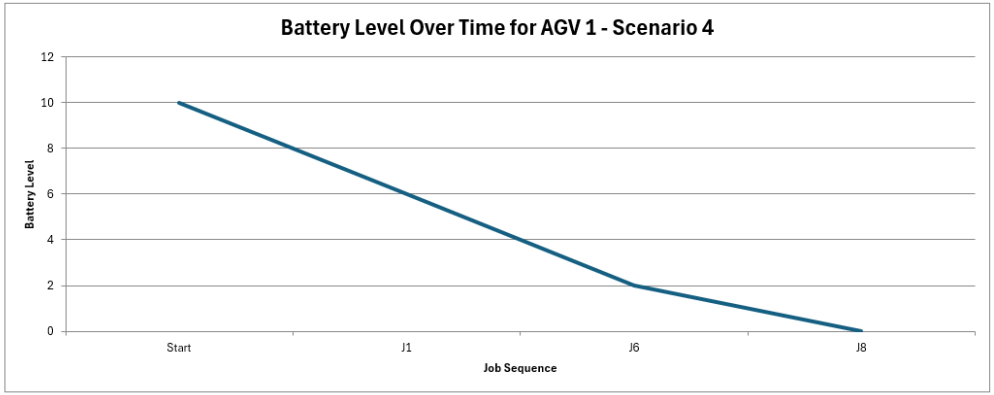
Charging Job	Job 1	Job 2	Job 3	Job 4	Job 5	Job 6	Job 7	Job 8	Job 9	Job 10
AGV 1	2	2	2	2	2	2	2	2	2	2
AGV 2	2	2	2	2	2	2	2	2	2	2
AGV 3	2	2	2	2	2	2	2	2	2	2
AGV 4	2	2	2	2	2	2	2	2	2	2
AGV 5	2	2	2	2	2	2	2	2	2	2

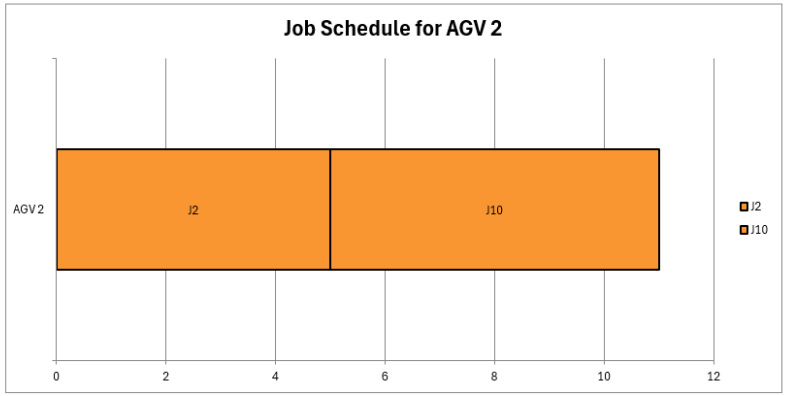
Energy Consu	Job 1	Job 2	Job 3	Job 4	Job 5	Job 6	Job 7	Job 8	Job 9	Job 10
AGV 1	4	5	3	6	3	4	6	2	8	5
AGV 2	4	5	3	6	3	4	6	2	8	5
AGV 3	4	5	3	6	3	4	6	2	8	5
AGV 4	4	5	3	6	3	4	6	2	8	5
AGV 5	4	5	3	6	3	4	6	2	8	5

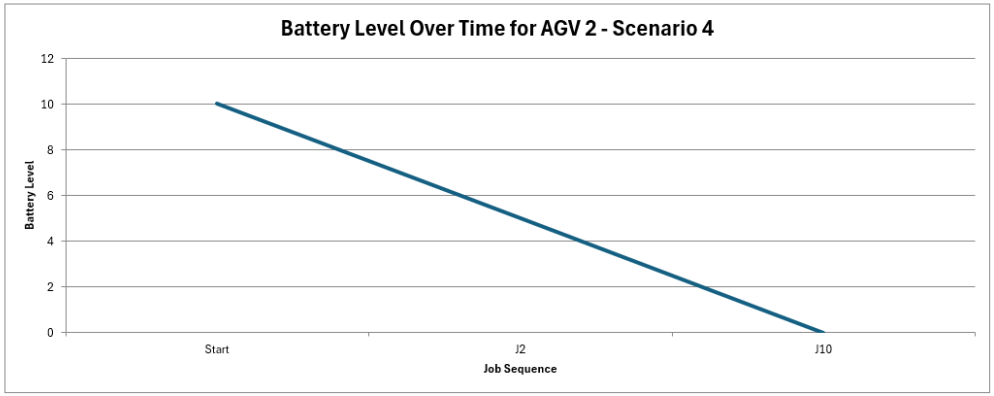
Battery Capacity per AGV	
AGV 1	10
AGV 2	10
AGV 3	8
AGV 4	8
AGV 5	6
Charging Tim	2
Completion T	11

Beispielfolie «Scenario 4, 2/5»

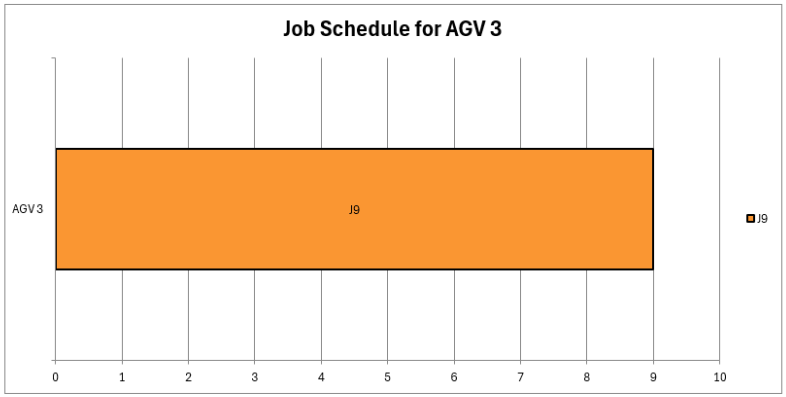


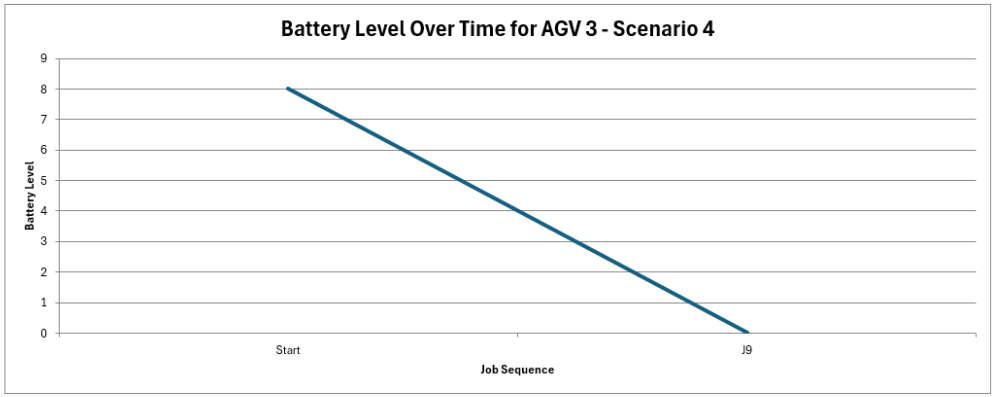


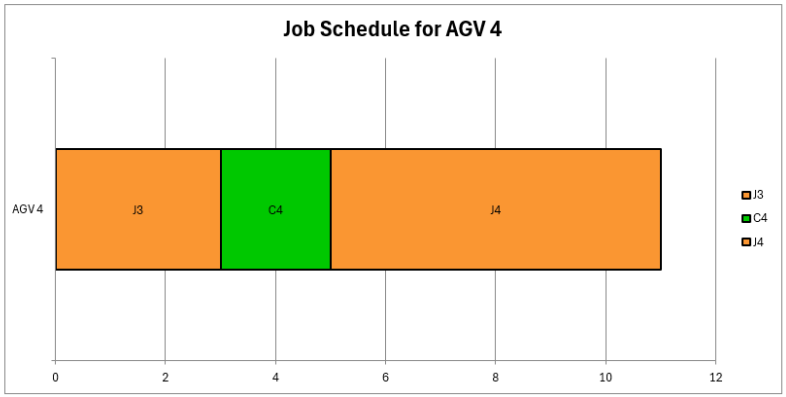


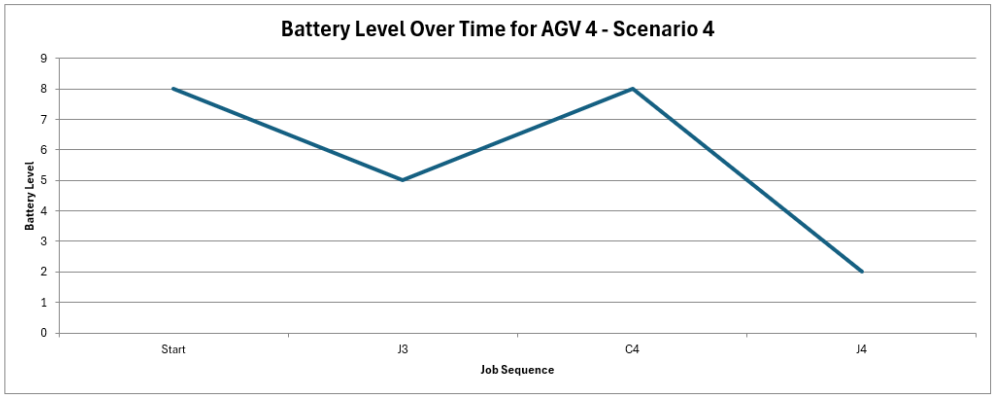


Beispielfolie «Scenario 4, 3/5»

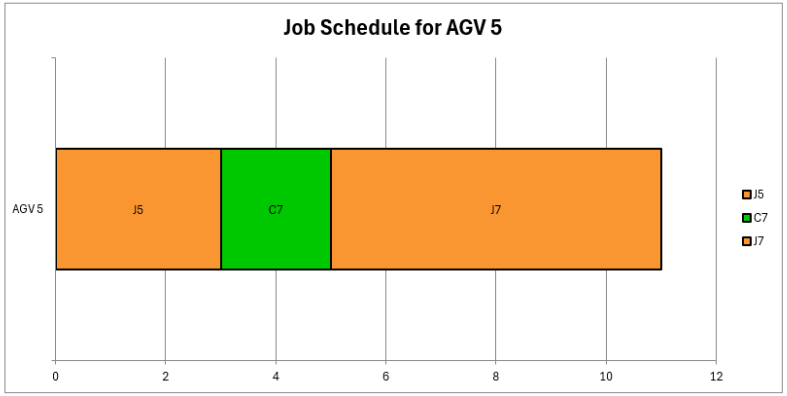


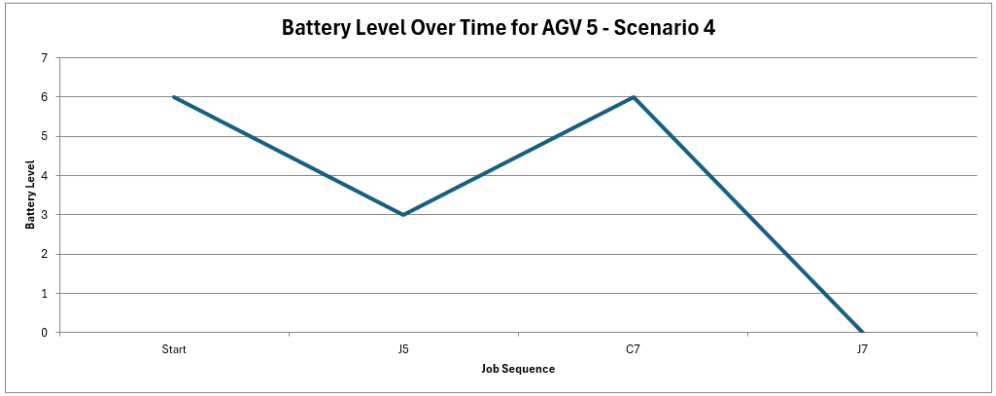
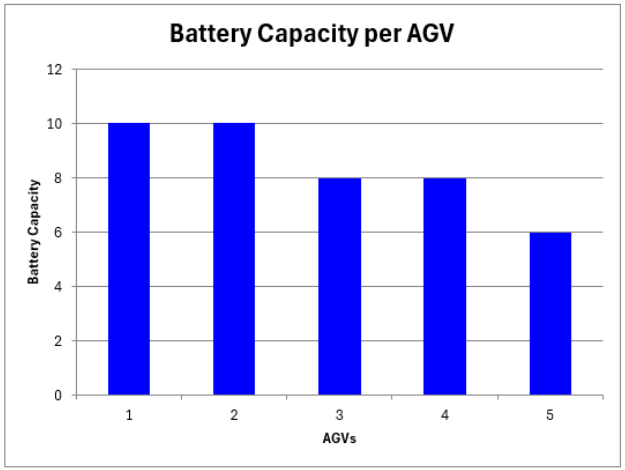
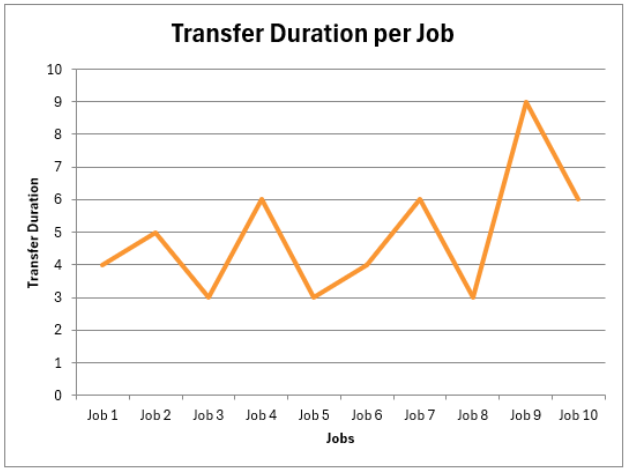
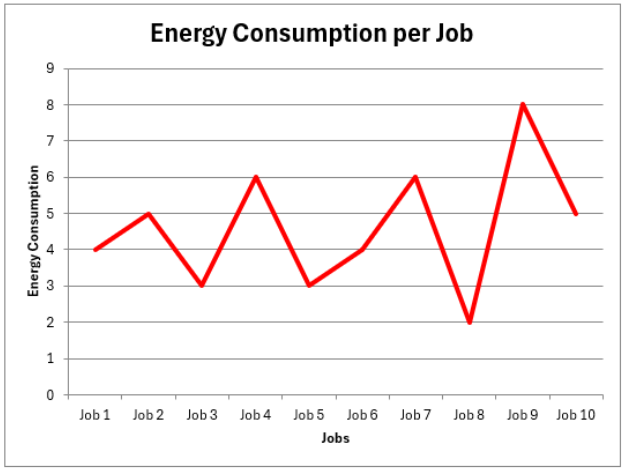
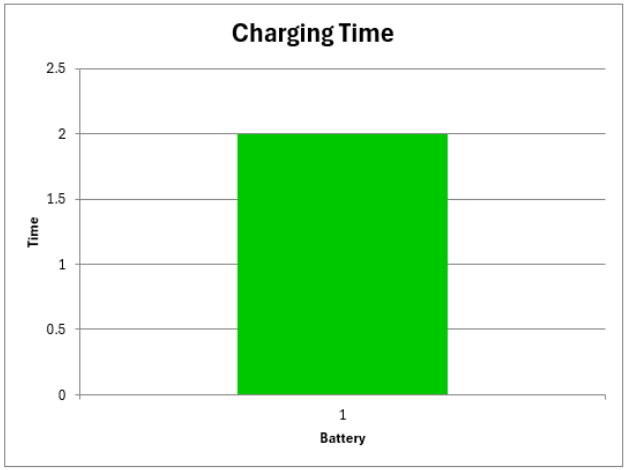






Beispielfolie «Scenario 4, 4/5»





Beispielfolie «Scenario 4, 5/5»